Proposed course · graduate and undergraduate

Applied Agentic AI

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Bring a real problem from your field, and learn to solve it with a team of AI agents that you direct.

You stay the human in the loop. AI should scaffold your work, not replace your judgment.

An interdisciplinary course to instruct students to utilize AI to help them reach new heights, whatever their background.

SCAFFOLD RESEARCH AND PERFORMANCE

1. small projects
   Everyday project work with agents
   Statistical analysis, project planning and scheduling.
2. small projects
   AI-simulated client
   You scope a problem with an AI-simulated client, the way you would with a real one.
3. Term project
   Build-and-verify team
   You build your own project. You set the goal, assign the work and check every result; the agents do the building.

HUMAN-IN-THE-LOOP

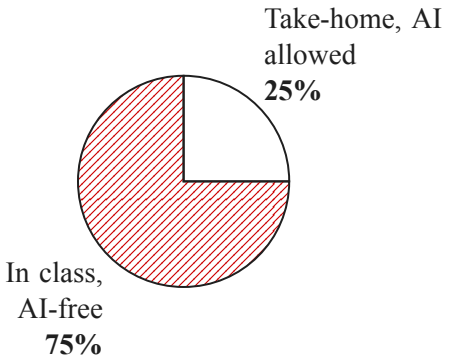
Try → Check → Record → Revise

STANDARDS AN AGENT MUST MEET

- **Rules** — Scripts the work must pass, trusted only after they fail
- **Regulations** — Guidance for judgment, given at every session start
- **Exemplars** — Earlier work to follow

Council of Agents
Different roles give different answers, even from one model

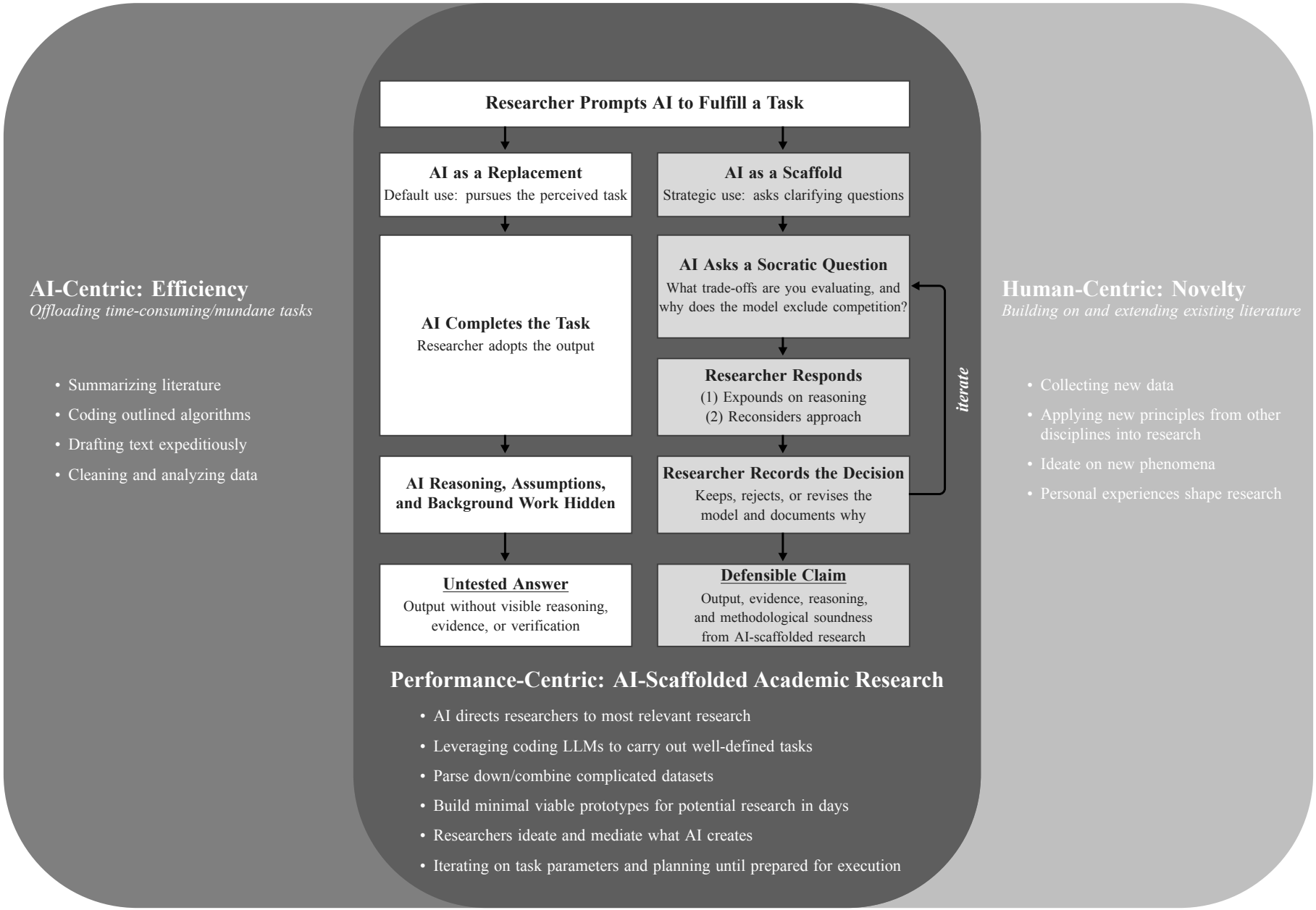
Share of course grade



- Students use any AI assistant or coding agent, such as Claude or Claude Code, ChatGPT or Codex, Google Gemini, or GitHub Copilot.
- No programming course or prior AI experience is required.

Syllabus
`jbenhart44.github.io/teaching/applied-agentic-ai-syllabus.pdf`

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The goal in this course is to teach students how to amplify their research and learning through applied AI.